1. No bus service.

All cases, including this one, have some aspect of randomness to them. As a result, you may find slightly different numbers during yours. This was the first case I made to ensure my model was working properly before implementing the bus routes.

Result:

- WALKED: 20000
- TOTAL TRAVEL TIME: 604665

The total number of person-minutes means the mean employee spent approximately 24 minutes walking to VMC station (perceived 30 minutes). Of course, some would probably take an Uber, but this assumes Uber is not an option.

2. The existing plan, AM rush hour:

- Existing route 20 - Jane runs every 13 minutes
- Existing route 320 - Jane Express runs every 14 minutes
- Rerouted route 21 - Vellore runs every 36 minutes along Edgeley to Langstaff.
- Rerouted route 26 - Maple runs every 38 minutes along the existing proposed route, i.e. along Creditstone north of Macintosh. Speed south of Creditstone / Locke assumed to be 17 km/h.
- New route 29 - Edgeley runs every 39 minutes along the existing Edgeley route of route 26 - Maple.

Result:

- Walked: 3864
- Route 20: 8477
- Route 320: 2368
- Route 29: 2705
- Route 21: 2122
- Route 26: 464
- Total Travel Time: 287050

I additionally tried scenarios where speeds were increased.

22 km/h result:

- Walked: 3831
- Route 20: 8434
- Route 320: 2343
- Route 29: 2703
- Route 21: 2121
- Route 26: 568
- Total Travel Time: 286238

30 km/h result:

- Walked: 3808
- Route 20: 8355
- Route 320: 2305

- Route 29: 2702
- Route 21: 2120
- Route 26: 710
- Total travel time: 285245

50 km/h result:

- Walked: 3784
- Route 20: 8241
- Route 320: 2271
- Route 29: 2699
- Route 21: 2117
- Route 26: 888
- Total travel time: 283913

There seems to be very little ridership on this route even when it is super fast. I imagine this is due both to the lack of area surrounding the route (being very close to Jane) and the Jane route running much more frequently.

The area seems to be a bigger factor, as a 50 km/h 26 would not surpass the 13-minute 20 - Jane within the study area unless it ran more than every 5 minutes (though interestingly, 320 ridership falls by nearly half, proportionally much more than 20 does, before this happens). With a 22 km/h assumption, this does not occur even when route 26 - Maple runs every minute on Creditstone.

Interestingly, there were nearly as many riders on route 21 (south of Langstaff). If I run similar cases with a more frequent 21, it surpasses route 29 at a 28-minute headway. Below is the result of such a case:

- Walked: 3661
- Route 20: 8264
- Route 320: 2303
- Route 29: 2648
- Route 21: 2654
- Route 26: 470
- Total travel time: 280831

3. The existing plan, but routes 26 and 29 both run on Edgeley. Frequencies kept the same.

Route 26's routing does not change (simply copy-paste the Edgeley distance data from 29 to 26). This service assumes no coordination between routes 26 and 29, and the chance of either bus coming when you need it to is entirely randomized. With 38 and 39 being so close to each other, during any rush hour there would not be very much variation in reality. However, if these numbers are inaccurate it may be a better measurement than one might think.

Result:

- Walked: 3549
- Route 20: 7749
- Route 320: 2213
- Route 29: 2488

- Route 21: 1921
- Route 26: 2080
- Total travel time: 264722

Route 26 took very little ridership away from route 29 and 21, in total less than 10% of both routes' ridership. Route 20 had the biggest drop, but also only did so around 10% perhaps because of a lack of service on Creditstone (it can be assumed the 464 Creditstone riders took route 20 - Jane instead).

4. Scenario 3 (AM rush), but the schedules of routes 26 and 29, both on Edgeley, are blended.

A blended schedule is one which allows for constant headways between routes or branches, e.g. two 30-minute routes combining to provide 15-minute service. The proposed service levels between routes 26 and 29 are very similar, so blending the schedules will provide very little difference in layover time. Both routes were thus treated as one route through the study area, running every 20 minutes (each branch every 40 minutes.

Result:

- Walked: 3417
- Route 20: 7468
- Route 320: 2143
- Routes 26 & 29: 5451
- Route 21: 1521
- Total travel time: 257189

This increases ridership of the two routes by nearly 20%. If both routes are improved until they have a 15-minute combined headway, their combined ridership would be expected to surpass that of route 20 - Jane whether or not route 21 - Vellore runs in the area.

5. Existing plan (AM rush), but instead of a new route on Creditstone one bus is added on Jane.

The frequency of route 20 - Jane during the AM rush would be improved from 13 to 12 minutes. This is a hypothetical scenario. It assumes route 29 ends at Vaughan Mills. This was primarily done to see if it were a more effective solution for reducing travel time than adding a new Creditstone route.

Result:

- Walked: 3894
- Route 20: 8971
- Route 320: 2283
- Route 26: 2882
- Route 21: 1970
- Total travel time: 284407

This plan increases the total number who walked, but decreased total travel time compared to the existing plan. Route 20 ridership went up compared to the existing plan by about as much as the route 26 ridership in the existing plan. It also took a small number of riders from route 320. The Edgeley and Vellore (21) services took riders from each other, partly due to randomness.

Additionally, I tried the same scenario but route 26 ends at Vaughan Mills, giving Edgeley a 39-minute frequency. Result:

- Walked: 3963
- Route 20: 9009
- Route 320: 2302
- Route 29: 2710
- Route 21: 2016
- Total travel time: 285755

The total ridership between routes 29 and 21 went down about 100 compared to the existing scenario, with those 100 either now walking or taking route 20 - Jane.

6. The previous two scenarios but without an improvement on 20 - Jane.

20 - Jane remains at its present 13-minute AM rush frequency. No Creditstone route is implemented. Route 21 - Vellore continues running along Edgeley south of Langstaff. The 2nd case has a 39-minute frequency on Edgeley.

Result 1:

- Walked: 4061
- Route 20: 8560
- Route 320: 2448
- Route 26: 2925
- Route 21: 2006
- Total travel time: 287690

Result 2:

- Walked: 4102
- Route 20: 8629
- Route 320: 2453
- Route 29: 2775
- Route 21: 2041
- Total travel time: 289132

This is a better comparison for scenario 2 (the existing plan). The one-minute change in frequency on Edgeley apparently made more of a difference than a Creditstone route.

7. The existing scenario, but one bus on 20 - Jane is replaced by one extra on 26 - Maple.

Route 20 - Jane frequency is decreased from 13 to 15 minutes, while 26 - Maple frequency is improved from 38 to 29 minutes. This would likely help people catching the GO Train more than it helps anyone in this business park.

Result:

- Walked: 4042
- Route 20: 7681
- Route 320: 2747
- Route 29: 2766
- ROUTE 21: 2164
- ROUTE 26: 600

- TOTAL TRAVEL TIME: 292115

20 - Jane ridership fell much more than 26 - Maple ridership rose. Clearly, this is not a good solution within the area.

8. Existing proposal for midday hours, taken at 1pm.

In the early afternoon, route 20 - Jane's frequency begins to decline for two hours before the PM rush. This scenario does not capture that. Route 20 - Jane runs every 15 minutes, 320 - Jane Express does every 17 minutes, and 26 - Maple runs on creditstone every 45 minutes.

Results:

- Walked: 6509
- Route 20: 9679
- Route 320: 3286
- Route 26: 526
- Total travel time: 377507

9. Case 8, but midday service is instead offered on Edgeley.

Same as case 8, but 26 - Maple instead runs every 45 minutes on Edgeley.

Results:

- Walked: 5748
- Route 20: 8576
- Route 320: 2954
- Route 26: 2722
- Total travel time: 338101

Case 9 shows a clear improvement over case 8. Total perceived travel time is down nearly 40,000 minutes compared to case 8 (an average of 2 minutes per person). This is bigger than the difference between the existing rush hour plan and modifying route 26 such that service on Edgeley operates on a 20-minute blended schedule. Clearly, this is a much better option.

I additionally looked at if route 320 ran its Summer 2026 midday schedule, which has increased service due to Canada's Wonderland's longer hours. The difference between these modified versions of cases 8 and 9 were similar. Route 320's ridership increased compared to the usual case primarily from people no longer walking (noticeably more than the number who took 320 instead of 20), which surprised me.

10. Instead of adding a new route, two buses are added to route 20 - Jane, increasing its frequency to every 12 minutes.

Results:

- Walked: 5983
- Route 20: 11302
- Route 320: 2715
- TOTAL TRAVEL TIME: 364983

There is a substantial decrease in the number of people who walked, but a much less substantial improvement in perceived travel time. This points to a very minor improvement for a majority of the population, who might barely notice the improvement. An Edgeley route still shows an improvement in both metrics compared to this one, so that solution is likely the best one to carry forward. However, it is more effective in both increasing ridership and reducing travel time than adding a Creditstone Road.

11. Existing midday service (no improvement).

Route 20 - Jane runs every 15 minutes, and route 320 - Jane Express runs every 17 minutes. There are no other bus routes in the area in this scenario.

Results:

- Walked: 6816
- Route 20: 9767
- Route 320: 3417
- Total travel time: 380754

The travel time difference is noticeably small, though most of the new route 26 riders are not

12. PM rush, existing plan (20 km/h assumption on Creditstone)

Route 20 - Jane runs every 15 minutes, but is much slower than during the AM rush. Route 26 - Maple runs every 41 minutes. Route 29 - Edgeley runs every 40 minutes. Route 21 - Vellore runs every 36 minutes. Route 320 - Jane Express runs every 17 minutes.

Results:

- Walked: 4689
- Route 20: 7797
- Route 320: 2591
- Route 29: 3059
- Route 21: 1342
- Route 26: 522
- Total travel time: 316236

Like during the AM rush, this version of Route 26 does not get much ridership within the business park. This points to investing the resources on Edgeley perhaps being a better option.

13. PM rush, but route 26 runs on Edgeley (no schedule coordination)

Results:

- Walked: 4301
- Route 20: 7172
- Route 320: 2405
- Route 29: 2826
- Route 21: 1112
- Route 26: 2184
- Total travel time: 291333

This is a decent travel time reduction. The travel time reduction here gives me medium (but not high) confidence that enhancing service on Edgeley may be the answer during rush hours as well.